

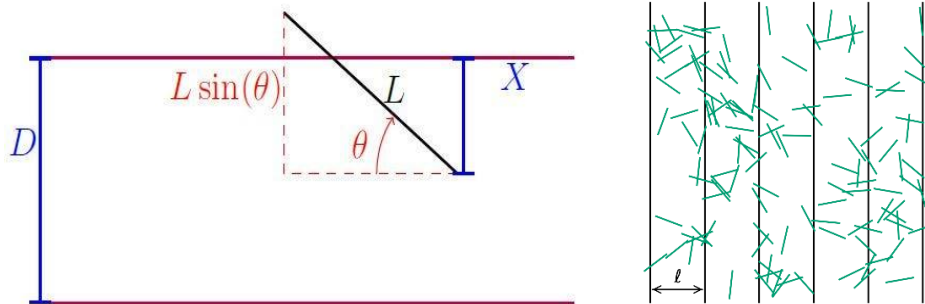
Testing the Uniformity of the Number of Crossings in Buffon's Needle Experiment

ENGS121 Mechanics Lab Section D

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1. Introduction

The experiment is done to test the behavior of random numerical data, the number of crossings in the Buffon's needle experiment. The experiment includes throwing a stack of 10 toothpicks per toss on a plane that has horizontal lines with the same distance between each other and finding the numbers of crossings for each stack of toothpicks for testing the uniformity of the number of crossings in a toss. The null hypothesis states that the probability for a crossing is $2/\pi$, while the alternative hypothesis states that the probability for a number of crossings to appear is uniformly distributed among tosses.



(Figure 1.1 Informative Diagrams of the Experiment)

2. Measurements

Procedure

After measuring the length of a toothpick series of horizontal lines on a paper with the same distance between has been drawn having the unit length of a toothpick in between them measured. A group of 10 toothpicks were thrown 40 times on the paper and the number of crossings per toss was recorded (See Table 1). Number of crossings squared per toss computed (See Table 2). Frequency and probability for each crossing category computed (See Table 3).

Degree of Freedom, Chi Square value and Standard Deviation computed and statistically evaluated.

Variables and Parameters

The response variable in this experiment is the **number of crossings** (a toothpick crossing the drawn line). The **angle** of the needle with respect to the parallel line and the perpendicular **distance** from the needle to the nearest line are variables as well since they change throughout the experiment, but we only need them for evaluating the theoretical part of the experiment. (See *Appendix 1*).

The **length of the toothpick** and the **distance between the parallel lines** are initially parameters, they stay constant during the whole experiment.

Recorded Measurements

Toss	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Crossings	8	3	6	8	7	7	7	6	5	6
Toss	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20
Crossings	3	9	4	4	6	6	7	10	4	6
Toss	#21	#22	#23	#24	#25	#26	#27	#28	#29	#30
Crossings	6	8	9	1	10	9	7	5	9	10
Toss	#31	#32	#33	#34	#35	#36	#37	#38	#39	#40
Crossings	6	9	6	9	8	7	8	4	4	4

(Figure 1.2 Number of crossings per toss)

Errors

Name	Type	Impact	Solution
Factory Imperfections	Systematic	Toothpicks length differences affect occurrence of a crossing	Toothpicks to be checked before tossing, and best fit toothpicks to be chosen
Biased Tossing	Systematic	When tossed in groups toothpicks often seem to fall next to each other creating a crowded and parallel not so random outcome.	Toothpicks to be dropped individually or more chaotic tossing style to be applied

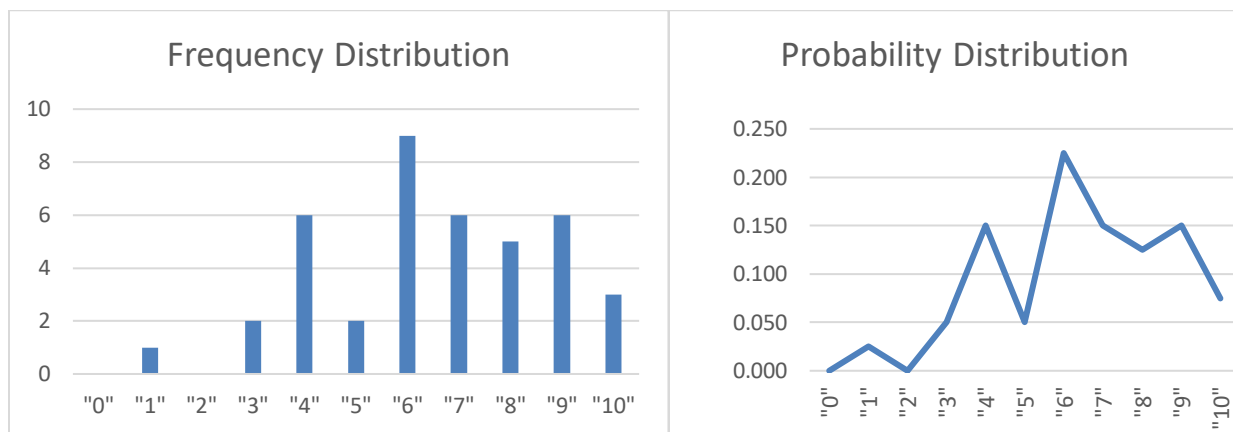
Instruments

Name	Quantity	Properties
Toothpick	10	Bamboo, ~65mm

(Figure 1.4 Probability distributions for first 6, 12, 21 tosses)

3. Data Analysis and Calculations

Determining whether the distribution is uniform or not



In both frequency and probability distribution diagrams it is shown that the number of crossings per toss is not uniformly distributed, and the observed data is crowded near the value “6”. Hence the distribution visually is seen to be **not uniform**.

Degree of Freedom

The Degree of Freedom can be calculated by knowing that the result of 10 cases can be used to calculate the remaining 11th one, hence **$df = \text{Number of categories} - 1 = 11 - 1 = 10$** .

Significance Level

$P_{df}(\chi^2) = 0.05$ meaning there is a 5% chance of rejecting the null hypothesis, from the Chi Square Distribution Table the critical data for one tailed test at **$df = 10$** and **$P_{df}(\chi^2) = 0.05$** can be found.

$$\chi_{0.05}^2 = 18.307$$

Chi Square Test

Cases	Obs. Freq. (O)	Obs. Prob.	Exp. Prob.	Exp. Freq. (E)	Chi Square
"0"	0	0.000	0.091	3.636	3.636
"1"	1	0.025	0.091	3.636	1.911
"2"	0	0.000	0.091	3.636	3.636
"3"	2	0.050	0.091	3.636	0.736
"4"	6	0.150	0.091	3.636	1.536
"5"	2	0.050	0.091	3.636	0.736
"6"	9	0.225	0.091	3.636	7.911
"7"	6	0.150	0.091	3.636	1.536
"8"	5	0.125	0.091	3.636	0.511
"9"	6	0.150	0.091	3.636	1.536
"10"	3	0.075	0.091	3.636	0.111
Sum	40	1	1	40	23.800

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i}$$

(Formula 1.1)

After applying the Chi Square Formula (See Formula 1.1) on values for number of crossings per toss the χ^2 value is calculated to be 23.800. Since χ^2 is greater than the critical value of $\chi^2_{0.05} = 18.307$, this means the observed data **does not follow a uniform distribution** at the chosen significance level.

Calculating the standard deviation

Toss	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
Crossings	64	9	36	64	49	49	49	36	25	36
Toss	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20
Crossings	9	81	16	16	36	36	49	100	16	36
Toss	#21	#22	#23	#24	#25	#26	#27	#28	#29	#30
Crossings	36	64	81	1	100	81	49	25	81	100

Toss	#31	#32	#33	#34	#35	#36	#37	#38	#39	#40
Crossings	36	81	36	81	64	49	64	16	16	16

(Figure 1.3 Number of crossings squared per toss)

4. Conclusion and Evaluation

The value $\chi^2 = 1.828$ is lower than the critical value of $\chi_{0.05}^2 = 11.070$, there is not enough evidence to support the alternative hypothesis, so the null hypothesis is reasonable at the 5% significance level.

In conclusion, the experiment was held to test whether the center of mass imbalance caused by the holes on die's surface makes it unfair. After measuring, recording, analyzing the observed data, it is concluded that the probabilities for each face to occur are equivalent and equal to $1/6$ and the holes on die's faces do not have a significant impact on it's performance.

5. References

Chi-Square distribution Table. (n.d.). In *Chi-Square Distribution Table* (p. 2).

<https://www.math.arizona.edu/~jwatkins/chi-square-table.pdf>